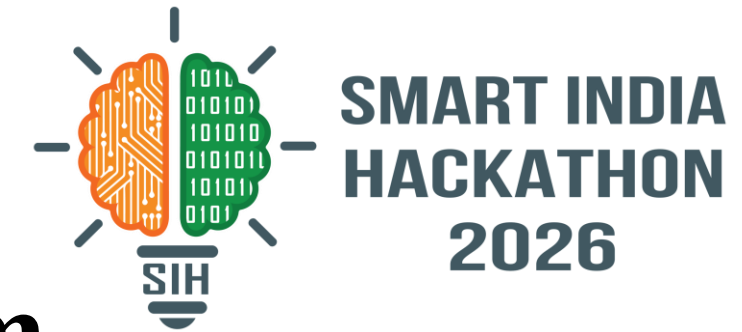


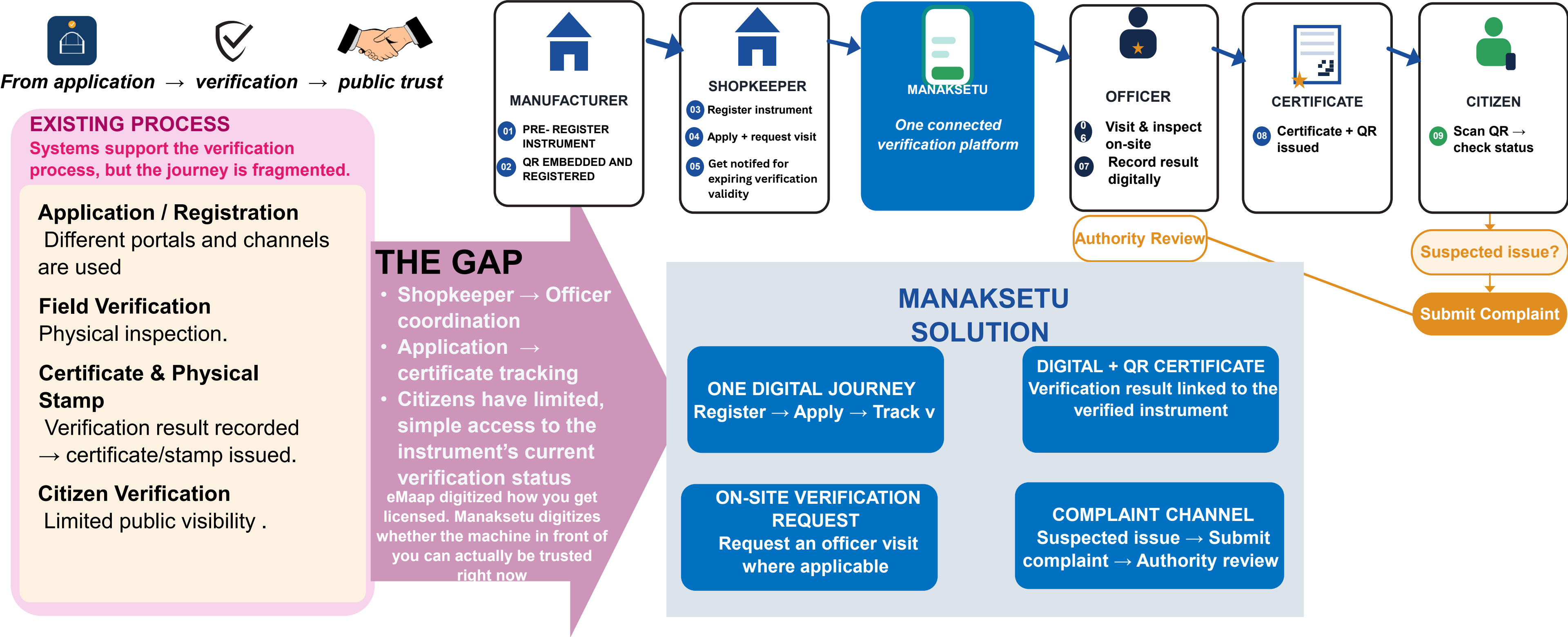
SMART INDIA HACKATHON 2026



MANAKSETU — Verification Integrity Platform for Weighing & Measuring Instruments

- **Problem Statement ID** – 26036
- **Problem Statement Title**- Online Verification System for Weighing & Measuring Instruments
- **Theme**- Transportation & Logistics
- **PS Category**- Software
- **Team ID**- 126
- **Team Name**- DevOps





**FRONTEND (User Interface)**

- React.js
- Tailwind CSS
- React Router
- Axios
- Responsive Design

**BACKEND**

- Node.js
- Express.js
- REST APIs
- File Handling
- Report Generation

**DATABASE**

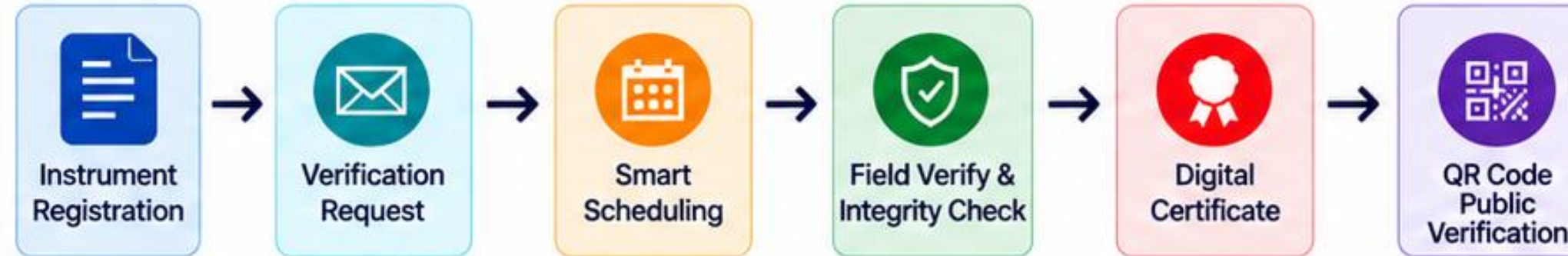
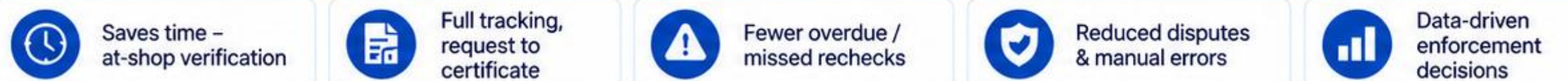
- PostgreSQL
- Secure & Scalable
- Data Storage

**QR / CERTIFICATE**

- Signed Hash
- Unique QR Code
- Public Verification

**DEPLOYMENT**

- Cloud (Render / Railway)
- Automatic Backups
- Scalable Infrastructure

CORE PLATFORM – END TO END FLOW**USERS****DATA & INTELLIGENCE LAYER****PUBLIC VERIFICATION****KEY BENEFITS****TECHNOLOGIES USED**

React.js



Tailwind CSS



Node.js



Express.js



PostgreSQL



QR Code

Cloud
(Render / Railway)



1. OPERATIONAL FEASIBILITY

BUILDS ON
EXISTING ECOSYSTEM

No New Workforce Required

Digitizes the exact duties LMOs and GATCs already perform.



Workflow-Aligned Roles

GATCs limited to verification scope, LMOs retain enforcement access.



Familiar Workflow

UI mirrors existing verification process for easier adoption.



Improves Officer Efficiency

Reduces coordination, follow-ups and manual workload.



2. CHALLENGES & MITIGATION

REAL CHALLENGES
PRACTICAL SOLUTIONS**Field connectivity gaps**
in rural / remote areas**Offline-first field app**
Works offline, auto-syncs when back online.**Officer adoption**
resistance to shift from paper**Phased rollout**
Familiar UI, training and handholding support.**Legacy record migration**
years of physical/isolated records**Fresh-forward start**
Allow gradual manual bulk-upload of legacy records.**State-wise process variation**
different rules and procedures**Configurable workflows**
Adaptable to state-specific requirements, phased onboarding.

3. SUSTAINABILITY MODEL

SELF-SUSTAINING
AND IMPACT ORIENTED

Nominal Vendor Processing Fee

Offsets logistics costs.

No charge to Customer / Manufacturer / LMO / Admin

GATC Subscription / Licensing

Platform tools on standard B2B model (private entities).



Better Penalty Tracking

Reduces government revenue leakage (statutory fines remain with the state treasury).



Growing Fee Volume

Lower compliance friction can increase verification adoption over time.



4. SCALABILITY & DEPLOYMENT

ONE PLATFORM. MANY STATES.
MANY INSTRUMENTS.

State-wise Multi-tenant Architecture

One platform, independently scoped per state — no redesign needed.



Instrument-type Agnostic

New categories (weighing, volumetric, length, etc.) can be added without re-architecting.



Pilot → Scale

Begin with selected districts, validate and expand gradually.



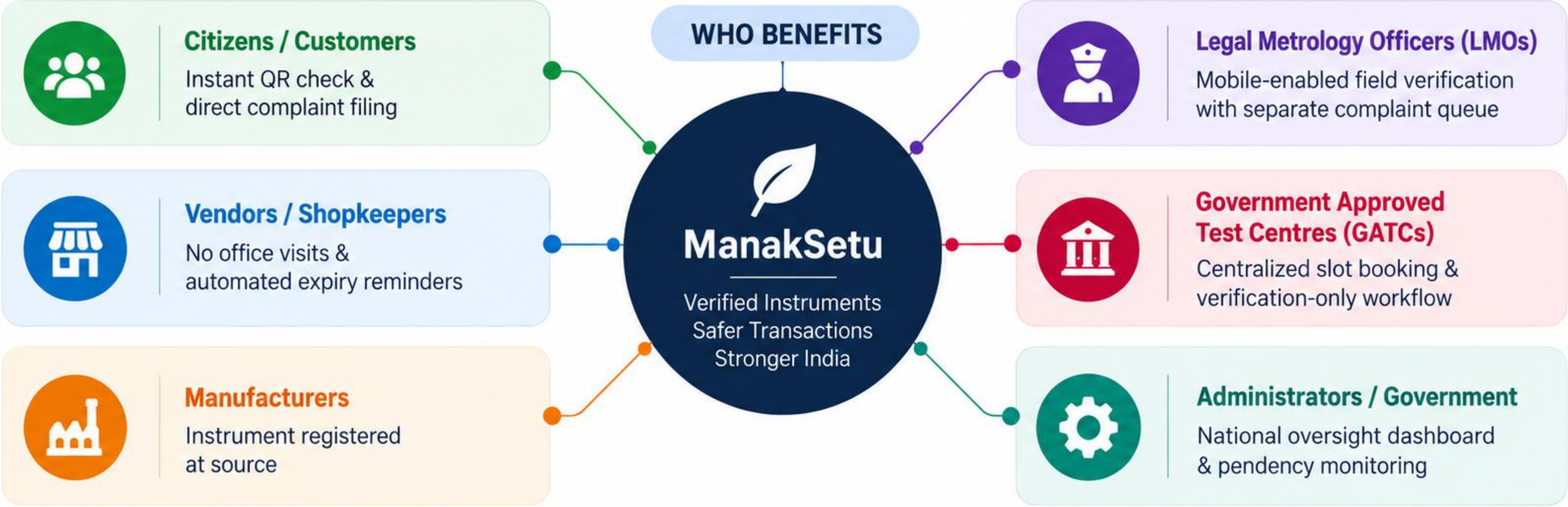
Integration-ready

Designed to coexist with existing state/national systems and databases.



READINESS SNAPSHOT

**Existing legal workflow**
Built on current framework**Minimal new infrastructure**
Software-based solution**Phased pilot ready**
Start small, scale confidently**Scalable across states**
Designed for national adoption



**OVERALL IMPACTS**

**Greater Transparency**
Trusted markets

**Higher Compliance**
Timely verification

**Reduced Disputes**
Fairer business environment

**Stronger Governance**
Data-driven decisions

Legal Metrology
Act, 2009
Section 24 — verification
& stamping powers

Legal Metrology
(General) Rules, 2011
Instrument tolerance
& specification norms

Legal Metrology
(GATC) Rules, 2013
Government Approved
Test Centre framework

State e-verification
portal reviewed
[emaap website](#)

Dataset / API
source
[dataset link provided](#)

PROTOTYPE PREVIEW — WIREFRAME

Main Dashboard

ManakSetu
LEGAL METROLOGY PORTAL

Home About How It Works Sign In / Demo

GOVERNMENT OF INDIA - LEGAL METROLOGY DIVISION

National Digital Legal Metrology Portal

Verify calibration, official government stamping, and physical lead seals on commercial weighing scales and measuring instruments.

Instant QR Verification Legal Metrology Act, 2009 Physical Lead Seal Verification

Portal Selection

ManakSetu
LEGAL METROLOGY PORTAL

Home About How It Works Sign In / Demo

PUBLIC ACCESS

Customer & Citizen Portal
Public Inspection & Grievance Box

Open Portal →

PRIVATE ACCESS

Manufacturer Portal
Phase 1 Hardware Registration

Sign In >

PRIVATE ACCESS

Business & Traders Portal
Phase 2 Purchase & Stamping

Sign In >

PRIVATE ACCESS

Legal Metrology Officer (LMO)
Field Inspection & Stamping

Sign In >

PRIVATE ACCESS

GATC Testing Laboratory
Lab Calibration & Pattern Approval

Sign In >

PRIVATE ACCESS

Central Administrator Console
Directory & Analytics

Sign In >

Citizen Verification (QR Scan)

ManakSetu
LEGAL METROLOGY PORTAL

Home About How It Works Sign In / Demo

Verify Any Instrument

Scan the QR code on the instrument or select a sample below to inspect its record.

System Active

Scan with Mobile

Point your phone camera at the QR code



Open Camera

Test Demo Instruments

Click any category to load its live verification record:

Grocery Scale 500 kg

Petrol Pump 45 L/min

Fabric Measure 1.0 m

Jewelry Scale 300 g

Diesel Dispenser 80 L/min

GOVERNMENT VERIFIED & ACTIVE

ABC-WM-500

Sharma General Store (Jaipur)

Project report

https://drive.google.com/drive/folders/1mVx_wvsKWdvZsCAWSG2UxHkMB-NQZAH